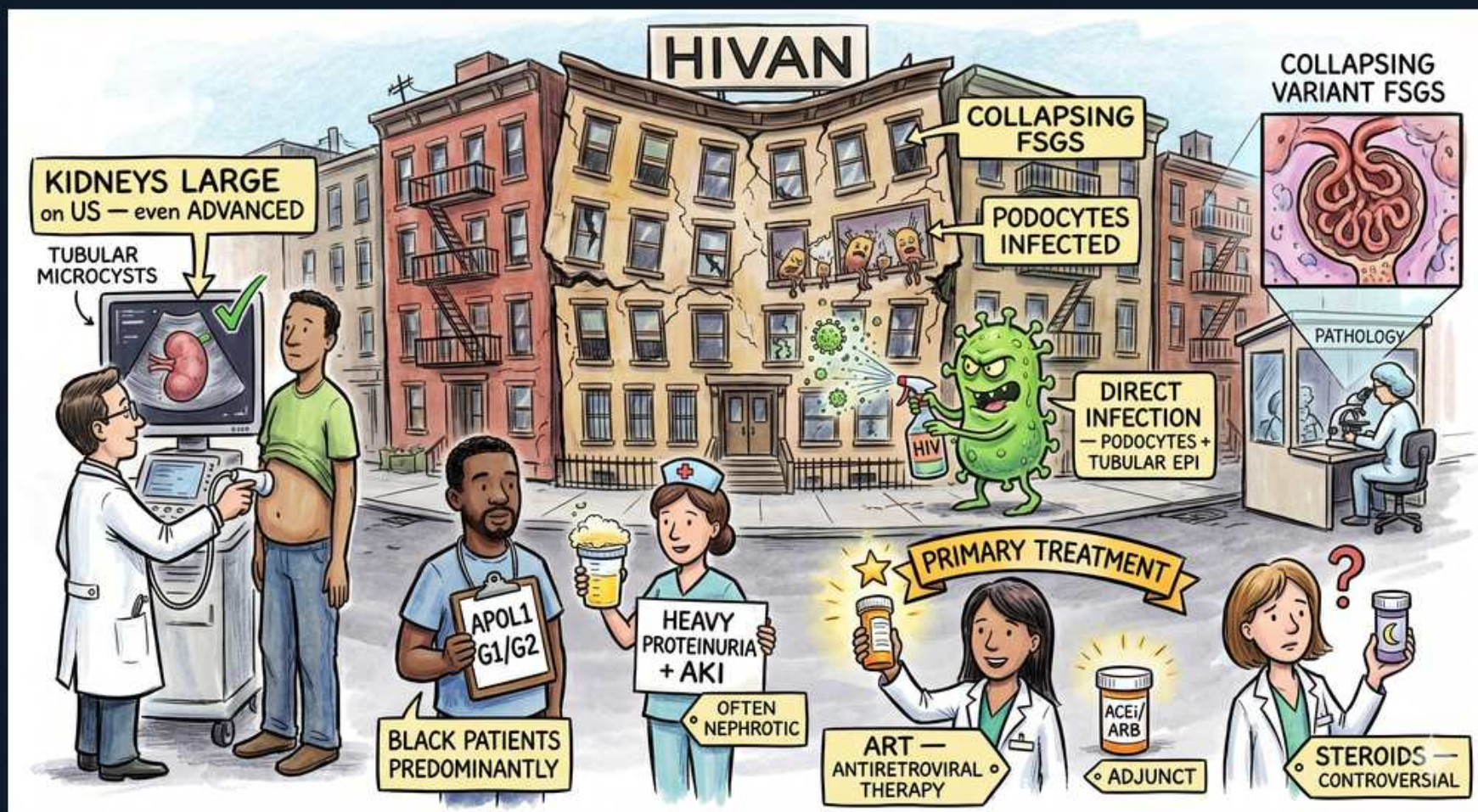


HIV-Associated Nephropathy (HIVAN) — Collapsing FSGS



1. Collapsing variant FSGS

WHAT YOU SEE

Pathology panel 'collapsing variant FSGS'

WHAT IT ENCODES

HIVAN is histologically a COLLAPSING form of focal segmental glomerulosclerosis: global collapse of the glomerular tuft with overlying podocyte hyperplasia/hypertrophy. It is the classic and most aggressive FSGS variant, often presenting with heavy nephrotic proteinuria and rapid GFR loss.

BOARD TRAP

HIVAN is the COLLAPSING variant of FSGS, not classic FSGS — collapse + podocyte proliferation distinguishes it and carries the worst prognosis.

2. Podocytes directly infected

WHAT YOU SEE

Green virus 'podocytes infected'

WHAT IT ENCODES

HIV directly infects glomerular podocytes and tubular epithelial cells, with viral gene products (e.g., nef, vpr) dysregulating the podocyte and driving dedifferentiation and proliferation. This direct infection mechanism differs from immune-complex GN.

BOARD TRAP

HIVAN injury is from DIRECT viral infection of podocytes/tubular cells — it is not an immune-complex deposition disease like HIV immune-complex kidney disease.

3. APOL1 G1/G2 risk alleles

WHAT YOU SEE

Man with cup 'APOL1 G1/G2'

WHAT IT ENCODES

Two APOL1 high-risk alleles (G1/G2), prevalent in people of West African ancestry, dramatically increase HIVAN risk. This genetic susceptibility explains the strong predominance in Black patients and overlaps with APOL1-related FSGS.

BOARD TRAP

APOL1 risk alleles (not HIV viral load alone) explain the Black-patient predominance — HIVAN is rare without APOL1 risk regardless of HIV status.

4. Black patients predominantly

WHAT YOU SEE

Caption 'Black patients predominantly'

WHAT IT ENCODES

HIVAN occurs almost exclusively in patients of African descent, reflecting the APOL1 genetic background. In other populations, HIV-related kidney disease more often takes other forms (immune-complex GN, thrombotic microangiopathy, drug toxicity).

BOARD TRAP

In non-African-ancestry HIV patients, suspect HIV immune-complex disease, TMA, or drug toxicity rather than classic HIVAN.

5. Large kidneys on US (even advanced)

WHAT YOU SEE

Ultrasound 'kidneys large on US even advanced'

WHAT IT ENCODES

Despite advanced renal failure, kidneys remain enlarged and echogenic on ultrasound, with tubular microcysts on histology. This preserved/large size is unusual because most chronic kidney disease shrinks the kidneys.

BOARD TRAP

Big echogenic kidneys in advanced renal failure is a HIVAN clue — most other causes of advanced CKD produce small, shrunken kidneys.

6. Tubular microcysts

WHAT YOU SEE

'tubular microcysts'

WHAT IT ENCODES

Tubulointerstitial injury with microcystic tubular dilation filled with proteinaceous casts is characteristic, contributing to the enlarged kidney size. Interstitial inflammation and edema accompany the glomerular collapse.

BOARD TRAP

Microcystic tubular dilation accompanies the glomerular collapse — HIVAN is a glomerular AND tubulointerstitial disease, not purely glomerular.

7. Heavy proteinuria + AKI (often nephrotic)

WHAT YOU SEE

Nurse cup 'heavy proteinuria + AKI, often nephrotic'

WHAT IT ENCODES

Classic presentation is nephrotic-range proteinuria with rapidly progressive renal failure, often in advanced/untreated HIV with low CD4. Untreated, it progresses to ESKD within months.

BOARD TRAP

HIVAN typically lacks hypertension and edema early despite massive proteinuria — rapid GFR decline with big kidneys, not classic nephrotic edema, is the tip-off.

8. ART — primary treatment

WHAT YOU SEE

Star 'primary treatment — ART antiretroviral therapy'

WHAT IT ENCODES

Antiretroviral therapy is the cornerstone treatment: suppressing HIV replication can stabilize or improve renal function and is indicated regardless of CD4 count when HIVAN is present. Early ART has markedly reduced HIVAN incidence.

BOARD TRAP

The first-line treatment of HIVAN is ART, not immunosuppression — controlling viral replication, not steroids, drives renal recovery.

9. ACEi/ARB adjunct

WHAT YOU SEE

Pill 'ACE/ARB adjunct'

WHAT IT ENCODES

ACE inhibitors or ARBs are adjuncts that reduce intraglomerular pressure and proteinuria, slowing progression alongside ART. They address the FSGS hemodynamic component common to all proteinuric glomerular disease.

BOARD TRAP

ACEi/ARB is adjunctive renoprotection — it complements but never substitutes for antiretroviral therapy in HIVAN.

10. Steroids — controversial

WHAT YOU SEE

Question mark 'steroids controversial'

WHAT IT ENCODES

Corticosteroids have been used in HIVAN with uncertain, controversial benefit and added infection risk in an immunocompromised host. They are not standard first-line therapy; ART and RAAS blockade are prioritized.

BOARD TRAP

Steroids are NOT standard HIVAN therapy — their benefit is unproven and they add infection risk; do not choose them over ART on the boards.

11. Direct infection of podocytes & tubular epi

WHAT YOU SEE

'direct infection — podocytes, tubular epi'

WHAT IT ENCODES

The combination of podocyte and tubular epithelial infection produces the dual glomerular collapse plus microcystic tubular injury that defines HIVAN's pathology, distinguishing it mechanistically from immune-complex HIV kidney disease.

BOARD TRAP

Collapsing FSGS + tubular microcysts from direct infection is HIVAN; granular immune deposits with low complement indicate HIV immune-complex GN instead.
